PROBLEM 40

Spring-Coupled Particles on a Vertical Axis and a Revolving Ring

Original statement · Problems.pdf, p. 28

Two Particles in a Rotating Frame

Two heavy particles P and Q, each of mass m, are connected with a spring of stiffness k as shown in the figure. An inertial reference frame S_0 is defined with origin O and vector basis $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$. Particle P slides without friction along the vertical axis, while particle Q slides without friction on a ring of radius R contained in the horizontal plane defined by O, $\mathbf{i}_0$ and $\mathbf{j}_0$. The center D of this ring is located a distance ℓ from O and rotates around the vertical axis of S_0 with a known constant angular rate Ω. A rotating reference frame S_1 is defined so that the $\mathbf{j}_1$ vector follows the position of D. We define the generalized coordinates ζ (vertical position of P) and ϕ (angle of Q on the circle from $\mathbf{j}_1$) as shown.

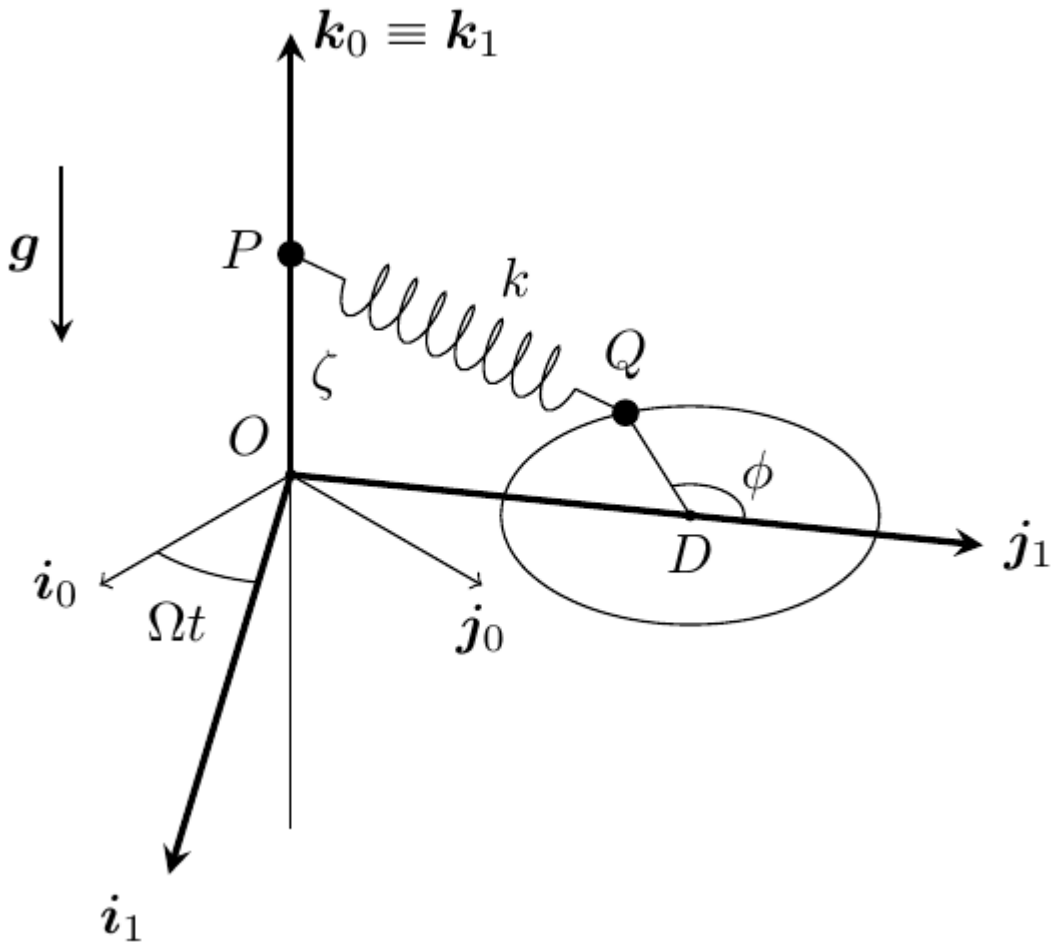


Figure: Particle P on the vertical axis k0 = k1, particle Q on a horizontal ring of centre D on the j1 axis, spring k between P and Q, frame S1 rotated by Omega t about k0 (Problems.pdf, p. 28).

(a) How many degrees of freedom does the system have? Define the vectors $\mathbf{i}_1, \mathbf{j}_1$ and $\mathbf{k}_1$ in terms of the vectors of $\mathcal{B}_0$.

(b) Write down the general expression of the position, velocity, and acceleration of P and Q with respect to S_1 in terms of ζ, ϕ and their derivatives.

(c) Identify and write down the general expression of all the forces acting on P and Q.

(d) Using S_1 for your analysis, write down the scalar equations of motion of P and Q.

(e) Starting from these equations, write down a system of two differential equations for ζ and ϕ, that do not depend on any other unknowns of the problem.

(f) Find one conservation law of the problem (i.e., a first integral of motion) in reference frame S_1.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** two heavy particles of mass m. P slides on the fixed vertical axis Oz_0; Q slides on a horizontal ring of radius R in the plane $z_0 = 0$ whose centre D, at distance ℓ from O, revolves about Oz_0 at constant rate Ω. A spring of stiffness k joins P and Q.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial, $\mathbf{k}_0$ vertical upward, gravity $-g\,\mathbf{k}_0$.
  2. Both contacts are smooth and bilateral: the reactions have no component along the direction in which each particle can move.
  3. The spring is linear and massless, with zero natural length (no natural length is given), so the force it exerts on P is $k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$ and on Q the opposite.
  4. The ring moves rigidly with S_1 (only its centre's motion is prescribed; it does not spin relative to S_1, which is irrelevant anyway for a smooth circular wire).
  5. ϕ is measured from $\mathbf{j}_1$ in the positive sense about $\mathbf{k}_1$ (from $\mathbf{j}_1$ towards $-\mathbf{i}_1$), as in the figure; this is also the convention of the official key, whose ring reaction contains the radial vector $-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1$.
- **Why S_1:** in S_1 the ring is fixed (centre at $\ell\,\mathbf{j}_1$), so Q moves on a fixed circle described by ϕ alone and P stays on the axis. The inertia forces (Notes Eq. 4.62) are the price; for P, on the rotation axis, they vanish.
- **Fixed (inertial) frame:** $S_0 = \{O; \mathcal{B}_0\}$, $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$.
- **Moving frame:** $S_1 = \{O; \mathcal{B}_1\}$ with the same origin, $\mathbf{k}_1 = \mathbf{k}_0$ and $\mathbf{j}_1$ pointing from O to D. S_1 is turned Ωt about $\mathbf{k}_0$ (the figure of the statement measures Ωt from $\mathbf{i}_0$ to $\mathbf{i}_1$), and we take $S_1 \equiv S_0$ at $t = 0$.
- **Ring basis:** $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{k}_1\}$ attached to Q: $\mathbf{e}_r$ from D to Q, $\mathbf{e}_\phi$ tangent to the ring; deduced in the next section.
- **Coordinates:** ζ, the height of P on the axis, and ϕ, the angle of Q on the ring.

Symbol	Meaning	SI unit
m	mass of each particle	kg

Symbol	Meaning	SI unit
k	spring stiffness	N/m
R	radius of the ring	m
ℓ	distance OD	m
Ω	constant angular rate of D	rad/s
g	gravitational acceleration	m/s^2
ζ, ϕ	height of P, angle of Q on the ring	m, rad
$N_1, \dots, N_4$	reaction components	N

(a) Degrees of freedom and the basis $\mathcal{B}_1$

> **Pedagogical Justification:** count coordinates minus independent constraints for each particle.

P is free in space (3 coordinates) but constrained to the vertical axis ($x = y = 0$): 1 DOF, the height ζ. Q (3 coordinates) is constrained to the plane $z_1 = 0$ and to the circle $\left|\mathbf{r}^Q - \overrightarrow{OD}\right| = R$: 1 DOF, the angle ϕ. The motion of D is prescribed, so it adds no unknown. Hence

$$\text{DOF} = 6 - 4 = 2, \qquad \text{generalized coordinates } (\zeta, \phi)$$

The vectors of $\mathcal{B}_1$ in terms of $\mathcal{B}_0$ are deduced in the next section.

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

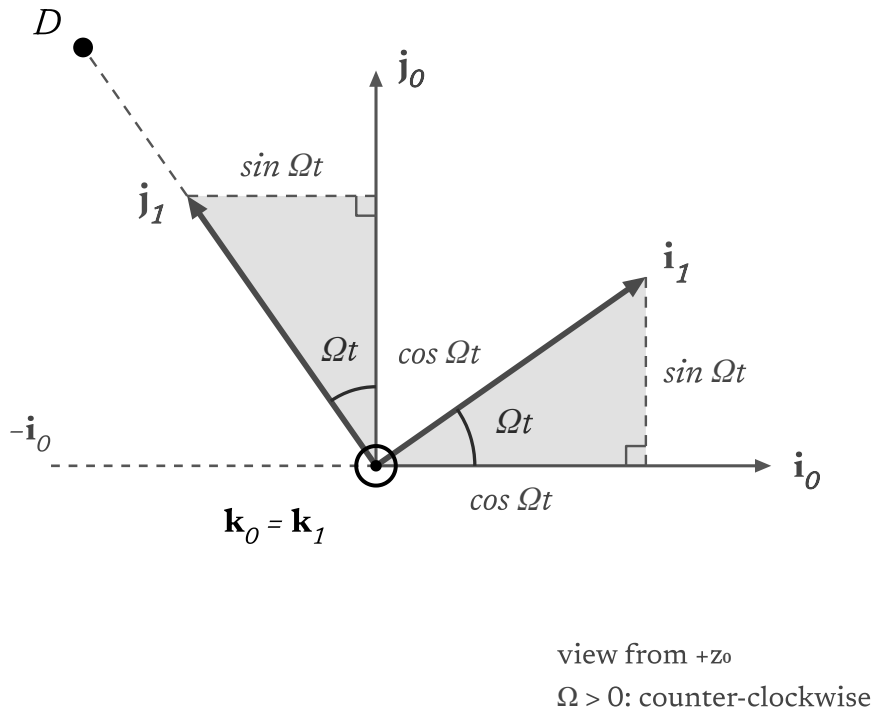


Figure 1. View from $+z_0$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$), so $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up. $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by Ωt, and $\mathbf{j}_1$ points from O to the centre D of the ring. $\mathbf{j}_1$ has legs $\cos \Omega t$ along $\mathbf{j}_0$ and $\sin \Omega t$ along $-\mathbf{i}_0$, hence its minus sign.

S_1 turns about the vertical, so we look down the axis from $+z_0$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper ($\odot$). With $\mathbf{i}_0$ to the right, $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up the page (right-hand rule: fingers from $\mathbf{k}_0$, towards the reader, to $\mathbf{i}_0$; thumb up). With $\Omega > 0$ the frame turns counter-clockwise in this view, and at time t it has turned Ωt:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the right triangle whose leg adjacent to Ωt lies along $\mathbf{i}_0$ ($\cos \Omega t$) and whose opposite leg lies along $+\mathbf{j}_0$ ($\sin \Omega t$).
- $\mathbf{j}_1$, the direction of D, starts on $\mathbf{j}_0$ and turns the same angle counter-clockwise, so it leans to the left: adjacent leg $\cos \Omega t$ along $\mathbf{j}_0$, opposite leg $\sin \Omega t$ along $-\mathbf{i}_0$. **The minus sign of $\mathbf{j}_1$ is that leftward lean.**
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \qquad \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \qquad \mathbf{k}_1 = \mathbf{k}_0$$

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\Omega t\cos\Omega t - (-\sin\Omega t)\sin\Omega t = \cos^2\Omega t + \sin^2\Omega t = 1$$

Orthogonality: in $[{}_0R_1][{}_0R_1]^T$ each diagonal entry is a squared row norm, $\cos^2\Omega t + \sin^2\Omega t = 1$ (or 1 for the third row), and the first two rows have dot product $\cos\Omega t\sin\Omega t - \sin\Omega t\cos\Omega t = 0$ (the third row is orthogonal to both), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$; simple rotation about $\mathbf{k}_0$ (Slide 16): $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$, and $\boldsymbol{\alpha}_{10} = \mathbf{0}$ because Ω and $\mathbf{k}_0$ are constant. Check with the chain rule: $\left.\frac{d\mathbf{j}_1}{dt}\right|_0 = -\Omega\cos\Omega t\,\mathbf{i}_0 - \Omega\sin\Omega t\,\mathbf{j}_0 = -\Omega\,\mathbf{i}_1 = \Omega\,\mathbf{k}_1 \times \mathbf{j}_1$.

The ring and the position of Q

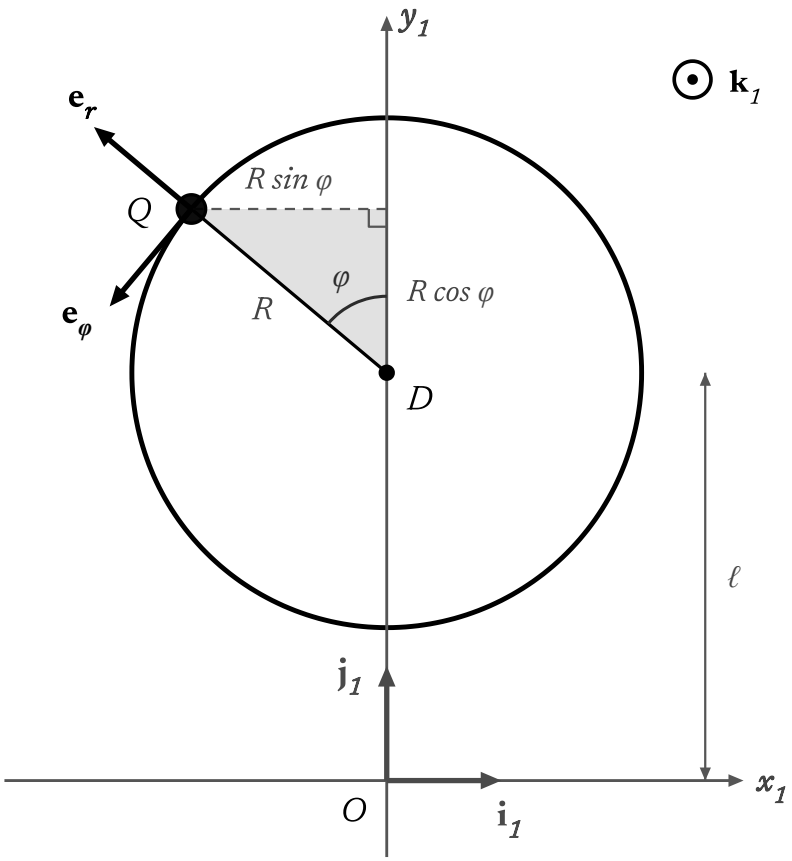


Figure 2. Plane of the ring, fixed in S_1, seen from above ($\mathbf{k}_1$ out of the paper). $OQ = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r$. In the shaded right triangle DFQ the hypotenuse is R, the leg along $\mathbf{j}_1$ is $R\cos\phi$ and the leg along $-\mathbf{i}_1$ is $R\sin\phi$. $\mathbf{i}_1, \mathbf{j}_1$ in blue, $\mathbf{e}_r, \mathbf{e}_\phi$ in orange.

The plane of the ring is fixed in S_1; we draw it from above ($\mathbf{k}_1$ out of the paper), with $\mathbf{i}_1$ to the right and $\mathbf{j}_1$ up, towards D. To go from O to Q we pass through the centre of the ring, because the distance from D is the one that stays constant:

$$\overrightarrow{OQ} = \overrightarrow{OD} + \overrightarrow{DQ}, \qquad \overrightarrow{OD} = \ell\,\mathbf{j}_1$$

$\overrightarrow{DQ}$ has length R and makes the angle ϕ with $\mathbf{j}_1$, measured counter-clockwise (towards $-\mathbf{i}_1$). Let F be the point of the line $D + s\,\mathbf{j}_1$ level with Q. In the shaded right triangle DFQ (right angle at F) the hypotenuse is $DQ = R$, the leg adjacent to ϕ is $DF = R\cos\phi$, along $+\mathbf{j}_1$, and the opposite leg is $FQ = R\sin\phi$, along $-\mathbf{i}_1$ because a counter-clockwise turn from $\mathbf{j}_1$ carries the point to the left. Hence the unit radial vector of the ring is

$$\mathbf{e}_r = \frac{\overrightarrow{DQ}}{R} = -\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1, \qquad \mathbf{r}_1^{Q} = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r = -R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1$$

Relation with the standard polar angle. The usual polar angle ψ about D is measured from $\mathbf{i}_1$, with $\mathbf{e}_r = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$. Matching requires $\cos\psi = -\sin\phi$ and $\sin\psi = \cos\phi$, which holds for $\psi = \phi + \frac{\pi}{2}$:

$$\cos\left(\phi + \tfrac{\pi}{2}\right) = \cos\phi\cos\tfrac{\pi}{2} - \sin\phi\sin\tfrac{\pi}{2} = -\sin\phi, \qquad \sin\left(\phi + \tfrac{\pi}{2}\right) = \sin\phi\cos\tfrac{\pi}{2} + \cos\phi\sin\tfrac{\pi}{2} = \cos\phi$$

(with $\cos\frac{\pi}{2} = 0$, $\sin\frac{\pi}{2} = 1$). So ϕ is the polar angle measured from $\mathbf{j}_1$, the direction $O \to D$; $\phi = 0$ is the point of the ring farthest from the axis and $\phi = \pi$ the nearest.

Ring basis. Differentiating $\mathbf{e}_r$ with respect to ϕ ($\frac{d}{d\phi}(-\sin\phi) = -\cos\phi$, $\frac{d}{d\phi}\cos\phi = -\sin\phi$) gives a vector of length $\sqrt{\cos^2\phi + \sin^2\phi} = 1$ tangent to the ring, pointing in the sense of increasing ϕ:

$$\mathbf{e}_\phi = \frac{d\mathbf{e}_r}{d\phi} = -\cos\phi\,\mathbf{i}_1 - \sin\phi\,\mathbf{j}_1, \qquad \mathbf{k}_1$$

Check: $\mathbf{e}_r \times \mathbf{e}_\phi = [(-\sin\phi)(-\sin\phi) - (\cos\phi)(-\cos\phi)]\,\mathbf{k}_1 = \mathbf{k}_1$, so $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{k}_1\}$ is right-handed and orthonormal ($\mathbf{e}_r \cdot \mathbf{e}_\phi = \sin\phi\cos\phi - \cos\phi\sin\phi = 0$). The inverse relations used below are $\mathbf{j}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi$ and $\mathbf{i}_1 = -\sin\phi\,\mathbf{e}_r - \cos\phi\,\mathbf{e}_\phi$.

Position of P. P is on the axis at height ζ: $\mathbf{r}_1^{P} = \zeta\,\mathbf{k}_1$. The spring vector is then

$$\mathbf{r}^{Q} - \mathbf{r}^{P} = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r - \zeta\,\mathbf{k}_1 = -R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1 - \zeta\,\mathbf{k}_1$$

Kinematics

(b) Relative kinematics of P and Q in S_1

Pedagogical Justification: O is the origin of S_1, so $\mathbf{r}_1 = \overrightarrow{OP}$ or $\overrightarrow{OQ}$ (Notes Eq. 3.16). The vectors $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are constant in S_1; $\mathbf{e}_r$ and $\mathbf{e}_\phi$ change only through $\phi(t)$, so the chain rule gives $\left.\dfrac{d\mathbf{e}_r}{dt}\right|_1 = \dfrac{d\mathbf{e}_r}{d\phi}\dot{\phi} = \dot{\phi}\,\mathbf{e}_\phi$ and $\left.\dfrac{d\mathbf{e}_\phi}{dt}\right|_1 = \dfrac{d\mathbf{e}_\phi}{d\phi}\dot{\phi} = \dot{\phi}\,(\sin\phi\,\mathbf{i}_1 - \cos\phi\,\mathbf{j}_1) = -\dot{\phi}\,\mathbf{e}_r$.

Particle P:

$$\mathbf{r}_1^P = \zeta\,\mathbf{k}_1, \qquad \mathbf{v}_1^P = \dot{\zeta}\,\mathbf{k}_1, \qquad \mathbf{a}_1^P = \ddot{\zeta}\,\mathbf{k}_1$$

Particle Q: with $\overrightarrow{OD} = \ell\,\mathbf{j}_1$ constant in S_1,

$$\mathbf{r}_1^Q = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r = -R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1$$

$$\mathbf{v}_1^Q = R\left.\frac{d\mathbf{e}_r}{dt}\right|_1 = R\dot{\phi}\,\mathbf{e}_\phi = -R\dot{\phi}\cos\phi\,\mathbf{i}_1 - R\dot{\phi}\sin\phi\,\mathbf{j}_1$$

By the product rule, $\mathbf{a}_1^Q = R\ddot{\phi}\,\mathbf{e}_\phi + R\dot{\phi}\left.\dfrac{d\mathbf{e}_\phi}{dt}\right|_1$:

$$\mathbf{a}_1^Q = R\ddot{\phi}\,\mathbf{e}_\phi - R\dot{\phi}^2\,\mathbf{e}_r$$

Forces and Equations of Motion

(c) Forces on P and Q (actual and inertia forces in S_1)

> **Pedagogical Justification:** in S_1 we apply $m\mathbf{a}_1 = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63) with $\mathbf{F}_i$ from Notes Eq. 4.62. Here $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$, so only the centrifugal and Coriolis forces survive (Slide 28).

Spring. With the spring vector of the change-of-basis section, the spring force on P is $\mathbf{F}_s^P = k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$ and on Q it is $\mathbf{F}_s^Q = -k\left(\mathbf{r}^Q - \mathbf{r}^P\right)$. In the ring basis, using $\mathbf{j}_1 = \cos\phi\,\mathbf{e}_r - \sin\phi\,\mathbf{e}_\phi$:

$$\mathbf{F}_s^Q = -k\left(\ell\cos\phi + R\right)\mathbf{e}_r + k\ell\sin\phi\,\mathbf{e}_\phi + k\zeta\,\mathbf{k}_1$$

Forces on P:

- Weight $-mg\,\mathbf{k}_1$.
- Reaction of the smooth axis, perpendicular to $\mathbf{k}_1$: $\mathbf{N}_P = N_1\,\mathbf{i}_1 + N_2\,\mathbf{j}_1$.
- Spring $\mathbf{F}_s^P = k\left[-R\sin\phi\,\mathbf{i}_1 + (\ell + R\cos\phi)\,\mathbf{j}_1 - \zeta\,\mathbf{k}_1\right]$.
- Inertia forces: P is on the rotation axis. $\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \Omega\zeta\,\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$, so $\mathbf{F}_C^P = \mathbf{0}$; $\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = \Omega\dot{\zeta}\,\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$, so $\mathbf{F}_{COR}^P = \mathbf{0}$.

Forces on Q:

- Weight $-mg\,\mathbf{k}_1$.
- Reaction of the smooth ring, perpendicular to the tangent $\mathbf{e}_\phi$: $\mathbf{N}_Q = N_3\,\mathbf{e}_r + N_4\,\mathbf{k}_1 = N_3\left(-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1\right) + N_4\,\mathbf{k}_1$.
- Spring $\mathbf{F}_s^Q$ above.
- Centrifugal force. Since $\mathbf{r}_1^Q$ has no $\mathbf{k}_1$ component, $\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^Q\right) = -\Omega^2\mathbf{r}_1^Q$ (from $\mathbf{k}_1 \times (\mathbf{k}_1 \times \mathbf{i}_1) = -\mathbf{i}_1$ and $\mathbf{k}_1 \times (\mathbf{k}_1 \times \mathbf{j}_1) = -\mathbf{j}_1$), hence

$$\mathbf{F}_C^Q = m\Omega^2\mathbf{r}_1^Q = m\Omega^2\left(\ell\,\mathbf{j}_1 + R\,\mathbf{e}_r\right) = m\Omega^2\left(\ell\cos\phi + R\right)\mathbf{e}_r - m\Omega^2\ell\sin\phi\,\mathbf{e}_\phi$$

- Coriolis force. From $\mathbf{e}_r \times \mathbf{e}_\phi = \mathbf{k}_1$ (cyclic: $\mathbf{e}_\phi \times \mathbf{k}_1 = \mathbf{e}_r$), $\mathbf{k}_1 \times \mathbf{e}_\phi = -\mathbf{e}_r$:

$$\mathbf{F}_{COR}^{Q} = -2m\Omega\,\mathbf{k}_1 \times R\dot{\phi}\,\mathbf{e}_\phi = 2m\Omega R\dot{\phi}\,\mathbf{e}_r$$

- Euler force and force due to the origin's acceleration: zero for both particles.

(d) Scalar equations of motion in S_1

> **Pedagogical Justification:** project $m\mathbf{a}_1 = \mathbf{F} + \mathbf{F}_i$ for each particle on an orthonormal basis adapted to its constraint: $\{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ for P and $\{\mathbf{e}_r, \mathbf{e}_\phi, \mathbf{k}_1\}$ for Q. This gives $3 + 3 = 6$ scalar equations for the 6 unknowns $\zeta, \phi, N_1, N_2, N_3, N_4$.

Particle P ($m\ddot{\zeta}\,\mathbf{k}_1 = -mg\,\mathbf{k}_1 + N_1\mathbf{i}_1 + N_2\mathbf{j}_1 + \mathbf{F}_s^P$):

$$\mathbf{i}_1: \quad 0 = N_1 - kR\sin\phi$$

$$\mathbf{j}_1: \quad 0 = N_2 + k\left(\ell + R\cos\phi\right)$$

$$\mathbf{k}_1: \quad m\ddot{\zeta} = -mg - k\zeta$$

Particle Q ($m\left(R\ddot{\phi}\,\mathbf{e}_\phi - R\dot{\phi}^2\mathbf{e}_r\right) = -mg\,\mathbf{k}_1 + N_3\mathbf{e}_r + N_4\mathbf{k}_1 + \mathbf{F}_s^Q + \mathbf{F}_C^Q + \mathbf{F}_{COR}^Q$):

$$\mathbf{e}_r: \quad -mR\dot{\phi}^2 = N_3 - k\left(\ell\cos\phi + R\right) + m\Omega^2\left(\ell\cos\phi + R\right) + 2m\Omega R\dot{\phi}$$

$$\mathbf{e}_\phi: \quad mR\ddot{\phi} = k\ell\sin\phi - m\Omega^2\ell\sin\phi$$

$$\mathbf{k}_1: \quad 0 = -mg + N_4 + k\zeta$$

Answers and Checks

(a) Degrees of freedom and $\mathcal{B}_1$

DOF $= 2$, coordinates (ζ, ϕ); $\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \mathbf{k}_1 = \mathbf{k}_0$.

(b) Kinematics in S_1

$\mathbf{r}_1^P = \zeta\,\mathbf{k}_1, \mathbf{v}_1^P = \dot\zeta\,\mathbf{k}_1, \mathbf{a}_1^P = \ddot\zeta\,\mathbf{k}_1; \mathbf{r}_1^Q = \ell\,\mathbf{j}_1 + R\,\mathbf{e}_r, \mathbf{v}_1^Q = R\dot\phi\,\mathbf{e}_\phi, \mathbf{a}_1^Q = R\ddot\phi\,\mathbf{e}_\phi - R\dot\phi^2\,\mathbf{e}_r$.

(c) Forces

As listed in the previous section: weights, reactions $\mathbf{N}_P = N_1\mathbf{i}_1 + N_2\mathbf{j}_1$ and $\mathbf{N}_Q = N_3\mathbf{e}_r + N_4\mathbf{k}_1$, spring forces $\pm k(\mathbf{r}^Q - \mathbf{r}^P)$, and on Q only the centrifugal force $m\Omega^2\mathbf{r}_1^Q$ and the Coriolis force $2m\Omega R\dot\phi\,\mathbf{e}_r$.

(d) Scalar equations of motion

The six projections of the previous section.

(e) Two equations free of the reactions

> **Pedagogical Justification:** the reactions appear only in the projections perpendicular to the allowed motions; the projections along the allowed motions ($\mathbf{k}_1$ for P, $\mathbf{e}_\phi$ for Q) are free of them.

Dividing the $\mathbf{k}_1$ equation of P and the $\mathbf{e}_\phi$ equation of Q by m and mR:

$$\boxed{\ddot\zeta + \frac{k}{m}\zeta = -g, \qquad \ddot\phi = \frac{\ell}{R}\left(\frac{k}{m} - \Omega^2\right)\sin\phi}$$

The other four equations then give the reactions once $\zeta(t), \phi(t)$ are known: $N_1 = kR\sin\phi$, $N_2 = -k\,(\ell + R\cos\phi)$, $N_3 = -mR\dot\phi^2 + \left(k - m\Omega^2\right)(R + \ell\cos\phi) - 2m\Omega R\dot\phi$, $N_4 = mg - k\zeta$. In particular the ring reaction has the form $\mathbf{N}_Q = N_3\,(-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1) + N_4\,\mathbf{k}_1$ of the official key.

The two equations are **decoupled**: the vertical spring component acts only on P's coordinate, and the horizontal one only on Q's, because the spring force is linear in $\mathbf{r}^Q - \mathbf{r}^P$ and P moves along $\mathbf{k}_1$ while Q moves in the plane $\perp \mathbf{k}_1$.

(f) A conservation law in S_1

> **Pedagogical Justification:** in S_1 the constraints (axis and ring) are fixed and smooth, so the reactions do no work; the Coriolis force on Q is perpendicular to $\mathbf{v}_1^Q$ (Slide 31); the weight and spring are conservative; and with $\boldsymbol{\omega}_{10}$ constant the centrifugal force has potential $-\tfrac{1}{2}m\Omega^2 r_\perp^2$ (Slide 30). Hence the mechanical energy of the system in S_1 is conserved (Slide 29).

Potential energies in S_1: weight of P, $mg\zeta$ (weight of Q: $mg\cdot 0$); spring $\tfrac{1}{2}k\left|\mathbf{r}^Q-\mathbf{r}^P\right|^2$; centrifugal on Q, $-\tfrac{1}{2}m\Omega^2\left|\mathbf{r}_1^Q\right|^2$ (here $r_\perp^Q=|\mathbf{r}_1^Q|$). With

$$\left|\mathbf{r}^Q-\mathbf{r}^P\right|^2 = R^2\sin^2\phi+(\ell+R\cos\phi)^2+\zeta^2 = \ell^2+R^2+2\ell R\cos\phi+\zeta^2, \qquad \left|\mathbf{r}_1^Q\right|^2=\ell^2+R^2+2\ell R\cos\phi$$

and dropping the constant $\tfrac{1}{2}\left(k-m\Omega^2\right)\left(\ell^2+R^2\right)$:

$$\boxed{E_1=\frac{1}{2}m\left(\dot\zeta^2+R^2\dot\phi^2\right)+mg\zeta+\frac{1}{2}k\zeta^2+\left(k-m\Omega^2\right)\ell R\cos\phi=\text{const}}$$

Check with the equations of part (e) (chain rule $\tfrac{d}{dt}\cos\phi=-\dot\phi\sin\phi$, $\tfrac{d}{dt}\zeta^2=2\zeta\dot\zeta$):

$$\frac{dE_1}{dt}=m\dot\zeta\ddot\zeta+mR^2\dot\phi\ddot\phi+mg\dot\zeta+k\zeta\dot\zeta-\left(k-m\Omega^2\right)\ell R\dot\phi\sin\phi=m\dot\zeta\left(\ddot\zeta+g+\frac{k}{m}\zeta\right)+mR\dot\phi\left(R\ddot\phi-\left(\frac{k}{m}-\Omega^2\right)\ell\sin\phi\right)=0$$

Because the equations are decoupled, each bracket vanishes separately, so in fact two first integrals exist: $\tfrac{1}{2}m\dot\zeta^2+mg\zeta+\tfrac{1}{2}k\zeta^2=\text{const}$ and $\tfrac{1}{2}mR^2\dot\phi^2+\left(k-m\Omega^2\right)\ell R\cos\phi=\text{const}$. The constants are fixed by initial conditions, which the statement does not give.

Checks

- **Particle P:** $\ddot\zeta+(k/m)\zeta=-g$ is a harmonic oscillator about $\zeta_{eq}=-mg/k$ with angular frequency $\sqrt{k/m}$, unaffected by the rotation (it lies on the axis, where centrifugal and Coriolis forces vanish).

- **Particle Q:** $\ddot{\phi} = \frac{\ell}{R}\left(\frac{k}{m} - \Omega^2\right)\sin\phi$ is a pendulum equation. The spring pulls Q towards the axis (towards $\phi = \pi$, the point of the ring nearest O) and the centrifugal force pushes it outward (towards $\phi = 0$). If $\Omega^2 < k/m$ the stable position is $\phi = \pi$; if $\Omega^2 > k/m$ it is $\phi = 0$; at $\Omega^2 = k/m$ the two effects cancel exactly and $\dot{\phi}$ is constant. The Coriolis force only loads the ring radially (N_3).
- **Limits:** $\Omega \to 0$ gives a fixed ring and $\ddot{\phi} = \frac{k\ell}{mR}\sin\phi$; $\ell \to 0$ (ring centred on the axis) gives $\ddot{\phi} = 0$, uniform motion, since every point of the ring is then equidistant from the axis. Geometry: $\phi = 0$ gives $\mathbf{r}_1^Q = (\ell + R)\,\mathbf{j}_1$, the far point of the ring, and $\phi = \pi$ gives $(\ell - R)\,\mathbf{j}_1$.
- **Dimensions:** $[k/m] = \mathrm{N/(m\,kg)} = \mathrm{s^{-2}}$, so $[(k/m)\zeta] = \mathrm{m/s^2} = [g] = [\ddot{\zeta}]$; $[(\ell/R)(k/m - \Omega^2)] = \mathrm{s^{-2}} = [\ddot{\phi}]$; $[N_3]$: $[mR\dot{\phi}^2] = [kR] = [m\Omega R\dot{\phi}] = \mathrm{N}$; $[E_1] = \mathrm{kg\,m^2/s^2} = [k\zeta^2] = [k\ell R] = \mathrm{J}$.
- **Symbolic check (sympy):** $\mathbf{a}_0$ of both particles obtained by differentiating $[{}_0R_1]\,\mathbf{r}_1$ twice matches Notes Eq. 3.27; the six scalar equations, the four reactions, and $dE_1/dt = 0$ along the motion are confirmed.
- **Check against the official key** (Problems.pdf, p. 76): "One of the forces acting on Q is the reaction of the ring, $\mathbf{N}_Q = N_3(-\sin\phi\,\mathbf{i}_1 + \cos\phi\,\mathbf{j}_1) + N_4\,\mathbf{k}_1$", which matches part (c).

Source: Problems.pdf (PDF page 28); key on PDF page 76